

Newton-Raphson Solver Application

User Guide

Overview

Interactive desktop application for visualizing and computing roots of mathematical functions using the Newton-Raphson numerical method.

Getting Started

1. Activate Python virtual environment
2. Run: `python -m app.main`
3. Application window opens automatically

Key Features

- Pre-loaded functions • Interactive selection • Real-time visualization
- Root-finding algorithm • Multi-language support • Export to CSV

How to Use

1. Select Function

Click dropdown and choose a mathematical function.

2. View Plot

Function plot displays automatically in the domain range.

3. Set Initial Guess

Enter initial value (x₀) near where function crosses x-axis.

4. Calculate Root

Click Calculate button. Result shows root and iteration count.

Input Parameters

Parameter	Description
Function	Select mathematical function
Initial Guess	Starting approximation (x ₀)
Tolerance	Convergence criterion

Tips for Best Results

- Initial guesses close to expected root = faster convergence
- Use plot to visually identify root locations
- Multiple roots may converge to nearest one

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